

OLIST E- COMMERCE PERFORMANCE ANALYSIS

A Power BI Business Intelligence Report

32558 – Business Intelligence

Table of Contents

Table of Contents	2
1. Introduction and Business Understanding	1
1.1 Business Context	1
1.2 Business Problem	1
1.3 Business Objectives	1
2. Data Acquisition and Loading	2
2.1 Dataset Source and Selection	2
2.2 Data Integration Process	2
2.3 Data Cleaning	3
2.4 Missing Values and Data Quality	5
3. Data Exploration and Profiling	7
3.1 Data Dictionary	7
3.2 Descriptive Statistics	8
4. Data Transformation and Wrangling	9
4.1 Data Type and Structure Cleaning	9
4.2 New Analysis Fields	10
4.3 Missing Values, Duplicates and Outliers	10
7. Dashboard Development and Visualisation Design	12
5.1 Dashboard Overview and KPI Cards	12
5.2 Visualisation Justification	12
5.3 Slicers and Interactivity	13
5.4 Dashboard Layout and Design	14
6. Interpretation and Insights	15
6.1 Audience-Tailored Communication	15
6.2 Key Findings	15
6.3 Real-World Implications	15
6.4 Recommendations	18
9. References	19

1. Introduction and Business Understanding

1.1 Business Context

The dashboard was created to provide a clear business view of Olist's sales, operations, customer satisfaction, and regional demand. The project examines Olist, a Brazilian e-commerce marketplace that connects customers, sellers, products, payments, reviews, and delivery services. There are several elements that lead to business success, such as sales performance, delivery time, freight cost, review scores, product categories, and customer locations. However, since the operational data are distributed across multiple CSV tables linking to all these elements, the fragmented structure of data has complicated holistic performance assessment, which gives itself the difficulty to evaluate the full business picture without integrating the data.

1.2 Business Problem

The core problem is the absence of a unified view of marketplace performance: sales trends are not readily visible, product revenue contributions are unclear, regional demand is uneven, and logistics factors such as delivery timeliness and freight costs may be degrading customer satisfaction. These limitations hinder data-driven decisions across marketing, inventory, and logistics, increasing the risk that high-value categories or high-demand regions are under-resourced while latent service issues remain undetected.

1.3 Business Objectives

To address these gaps, the team aims to construct an integrated Power BI dashboard to consolidate the multi-table dataset, enable monthly revenue and order trend analysis, identify top revenue-generating categories, map customer orders by Brazilian state, and monitor delivery status, freight ratios, and review scores. The explicit objective to "Build an integrated Power BI dashboard" frames the analytical work and subsequent recommendations for product strategy, logistics optimisation, and customer-experience improvements.

2. Data Acquisition and Loading

2.1 Dataset Source and Selection

Olist is a Brazilian e-commerce platform where sellers and buyers around Brazil make trade in the platform. On Kaggle, they have provided a dataset containing around 100k orders from 2016 to 2018 made at multiple marketplaces within Brazil. Olist connects small businesses from all over Brazil to channels without hassle and with a single contract. Merchants are able to sell their products through the Olist Store and ship them directly to the customers using Olist logistics partners. After a customer purchases the product from Olist Store a seller gets notified to fulfill that order. Once the customer receives the product, or the estimated delivery date is due, the customer gets a satisfaction survey via email where they can give a verdict for the purchase experience and write down some comments (Olist et al, 2018).

The dataset itself includes a lot of information we can use to make various KPI's within it. It contains eight separate .csv files which needs to be joined together to make sense of the information. The information contains sales, products, delivery, freight pricing, payments, customers, sellers, and their locations, making this a realistic scenario for an e-commerce platform.

2.2 Data Integration Process

The Olist CSV files were joined using Merge Queries in Power BI Power Query. For the Olist dataset, Join was chosen instead of Union because the tables had different structures but shared common keys. Union is used when datasets have the same columns and need to be stacked vertically, while Join was more appropriate here because each Olist table contained different information about the same marketplace transactions, such as orders, customers, products, sellers, freight, delivery, and reviews.

The integration started with olist_orders_dataset as the main table because it contained the core order information, including order_id, order status, purchase date, and delivery dates. The customer, order item, product, translation, seller, and review tables were then joined through their relevant keys. This created a final Olist_Final_Analysis table that connected sales value, product category, freight cost, delivery performance, review score, and customer location for dashboard analysis.

Table 1. Data Integration Process in Power BI Power Query

Original table	Join key	Purpose of integration
olist_orders_dataset	Base table	Contains order status, purchase date, and delivery date fields
olist_customers_dataset	customer_id	Adds customer city and state for regional demand analysis
olist_order_items_dataset	order_id	Adds product price, freight value, product

		ID, seller ID, and item-level details
olist_products_dataset	product_id	Adds product category and product attributes
product_category_name_translation	product_category_name	Adds English product category names for clearer reporting
olist_sellers_dataset	seller_id	Adds seller city and state information
olist_order_reviews_dataset	order_id	Adds customer review scores and review information
Olist_Final_Analysis	Final output	Combined dataset used for sales, delivery, freight, satisfaction, product, and regional analysis

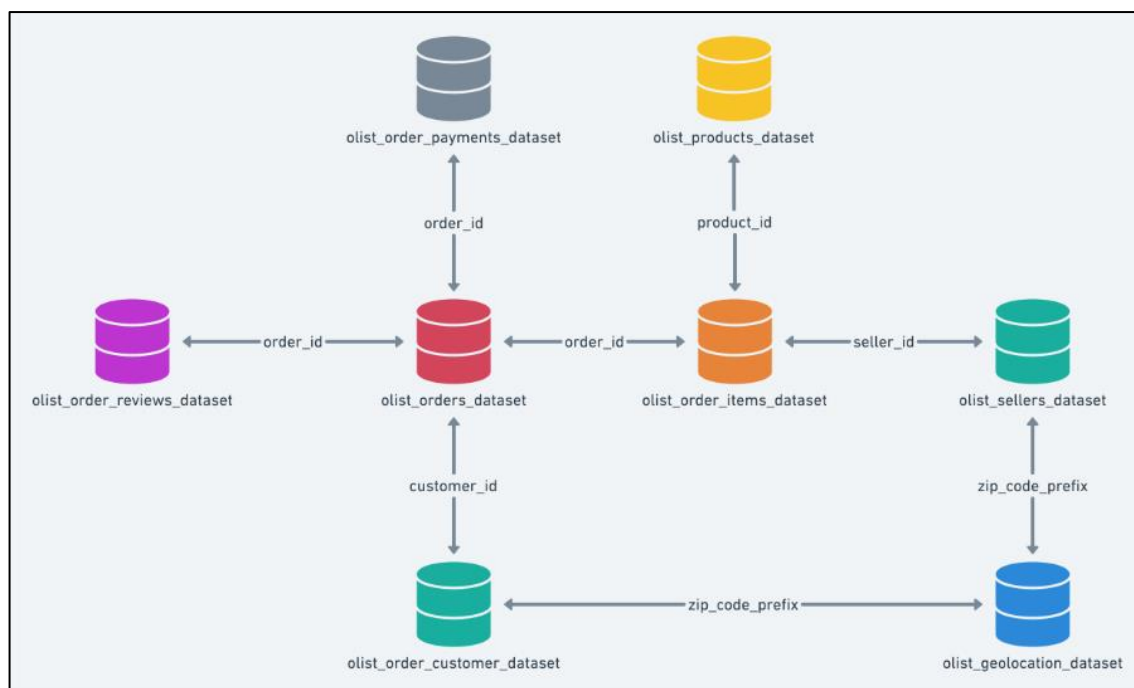


Figure 1. Original Olist Dataset Schema Used to Guide the Integration Process

2.3 Data Cleaning

Once we have successfully integrated the dataset with Power BI, the next step is to ensure that the data lacks duplicated values, is clean, easily readable and filters blank/NULL values.

Dupe-checking can be done by using a specific command in Power BI, `Table.Distinct(Source)`, onto the main unified dataset. This is to only be done once and ensure that there are no duplicated rows involved. Thankfully, there were no duplicates detected with this command.

One problem with the dataset is its lack of cleanliness; values are often stuck in lowercase, has underscores around the value, and are also stuck in Portuguese. A column called “Product_Category” was stuck in Portuguese, and therefore needed a direct translation. While the dataset provided a translation for “Product_Category”, the aforementioned underscores and lowercase letters remained, so we fixed them to improve readability.

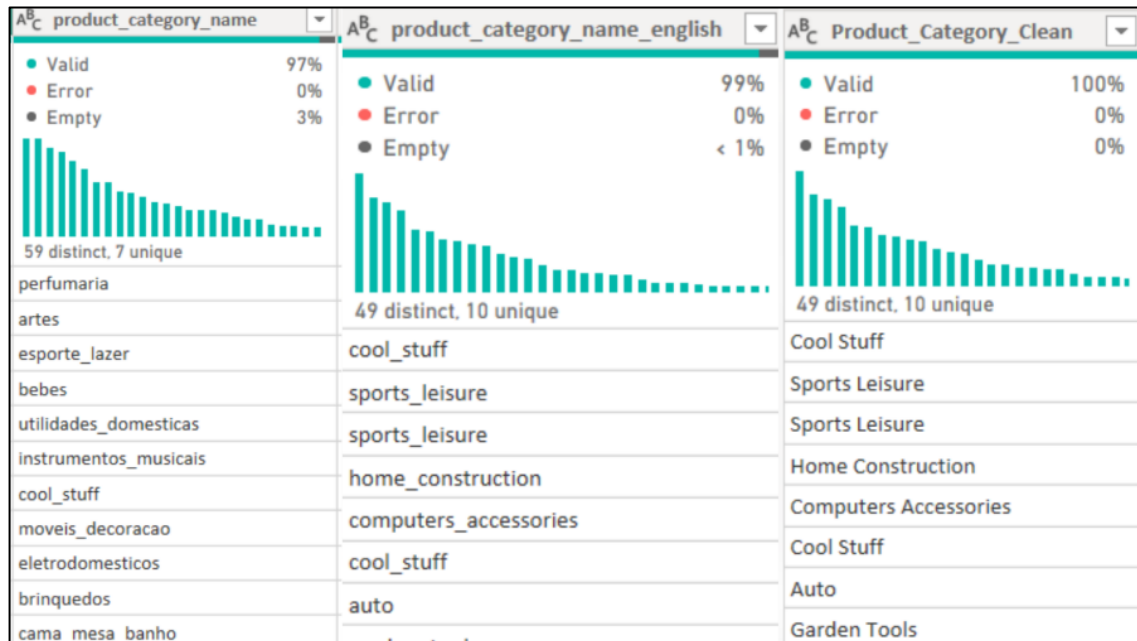


Figure 2. Product Category names in Portugese, English, and Cleaned Formatting

Another problem concerns with outliers. As a marketplace, anomalous data may exist that far exceeds the expected values, so an entire new series of columns were made to help identify the quantity of rows that were struck with a value too large (or too small). More columns designed to focus on this specific aspect were also made to help with blank/NULL values.

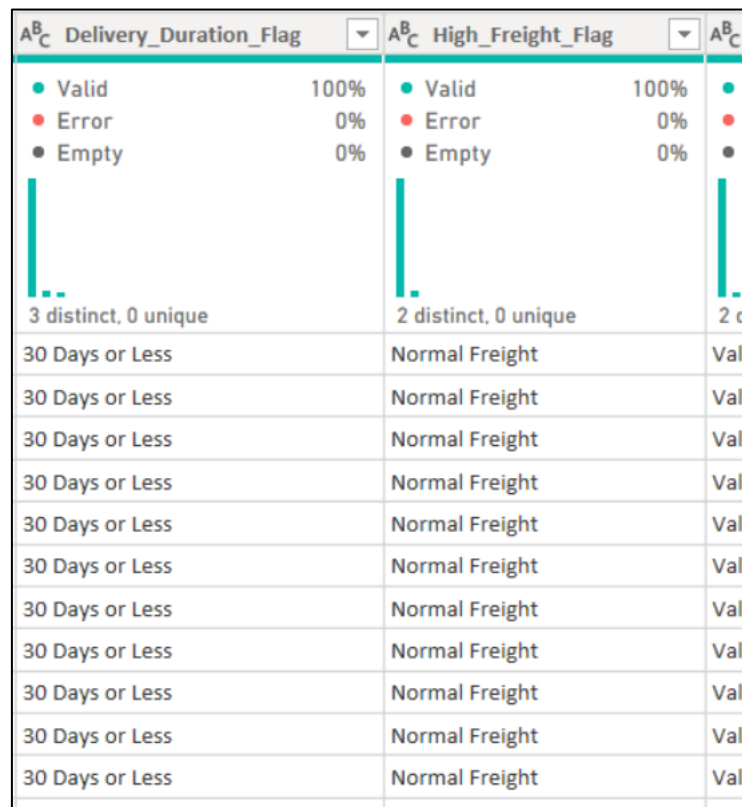


Figure 3. Flags to determine duration outlier and freight outlier

2.4 Missing Values and Data Quality

Within the dataset, some values aren't as valuable as others. Additionally, blank and null values may exist, causing problems for that dataset. With this in mind, some of them require special handling that may result in their removal. Listed in the table below are the measures taken to ensure that all fields are useful.

Table 2. Data Quality handling table

Field	Issues	Reason(s)	Handling
order_delivered_customer_date	Some blank/NULL data	Order may be cancelled, not shipped, not delivered or unavailable	Flagged using Delivery_Date_Status
Delivery_Days	Some blanks	order_estimated_delivery_date doesn't exist, or the order doesn't ship	Flagged with Delivery_Data_Quality
Delay_Days	Some blanks	order_estimated_delivery_date doesn't exist, delay cannot be calculated	Use Delivery_Status to classify these records

review_score	Some blanks	Customer didn't leave review sometimes	Use Review_Status to determine if user reviewed product
Product_Category_Clean	Missing categories	Source table contains NULL values	Replace NULL values with "Unknown Category"
review_comment_title / review_comment_message	Mostly blank and in Portuguese	Optional free-text comments were not suitable for structured BI analysis	Removed from final dashboard dataset

3. Data Exploration and Profiling

3.1 Data Dictionary

The final Olist_Final_Analysis table contains 114,092 rows at order-item level.

Table 3. Data Dictionary of the important datasets.

Field	Type	Description	Dashboard use
order_id	Text	Unique order identifier. Repeated values can appear because the final table is at order-item level.	Total orders and joins
order_purchase_timestamp	Date/Time	Date and time when the order was placed.	Monthly revenue and order trend
price	Decimal	Product item price before freight.	Total revenue and product revenue analysis
freight_value	Decimal	Freight cost charged for the item.	Total freight cost and freight ratio analysis
Product_Category_Clean	Text	Cleaned English product category name.	Product category chart and slicer
review_score	Whole Number (Integer)	Customer review score from 1 to 5.	Average review score and satisfaction analysis
customer_state	Text	Customer's Brazilian state code.	Regional demand and map analysis
Delivery_Days	Number (Float)	Number of days between purchase and actual delivery.	Average delivery days
Delay_Days	Number (Float)	Difference between actual and estimated delivery date. Negative values mean early delivery.	Delivery status classification
Delivery_Status	Text	Groups delivered orders as Early, On Time, or Late.	Delivery performance chart
Freight_Ratio	Decimal / %	Freight cost divided by product price.	Freight and satisfaction scatter chart
Purchase_Year Month	Text / period	Month extracted from purchase date.	Monthly trend chart

On Table 3, the most important tables are listed. These are generally considered the most important fields where they often appear.

It's worth noting that the actual data dictionary is significantly longer and has more fields than Table 3. The full dictionary will be viewable in Appendix A.

3.2 Descriptive Statistics

Field	Sum	Profile Mean	Profile Median	Profile Min	Profile Max	Profile Std Dev	Profile Blank %	Interpretation
Delay_Days		-11.11	-12.00	-146.00	188.00	10.00	2.85%	Identify early/late delivery
Delivery_Days		12.01	10.00	0.00	209.00	9.45	2.85%	Measures fulfillment duration
Freight_Ratio		0.32	0.23	0.00	26.24	0.35	0.68%	Identify high freight cost case
freight_value	2,263,949	19.98	16.26	0.00	409.68	15.78	0.68%	Assess shipping cost
Price	13,651,923	120.48	74.90	0.85	6,735.00	183.28	0.68%	Shows item revenue spread
Review_Score		4.02	5.00	1.00	5.00	1.40	0.84%	Measures consumer satisfaction

Figure 4. Table containing the sum, mean, median, min, max, standard deviation and blank/null values.

Figure 4 summarises key numeric fields using mean, median, minimum, maximum, standard deviation, blank percentage, and relevant totals. The average Delay_Days value is approximately -11 days, meaning delivered orders were often earlier than the estimated delivery date. However, the minimum and maximum values show that some orders were delivered much earlier or much later than expected, supporting the need to flag delivery-duration outliers.

4. Data Transformation and Wrangling

4.1 Data Type and Structure Cleaning

The raw CSV table names were standardised in Power BI Power Query to make the data model easier to understand.

Table 3. *Before-and-After Table Name Standardisation in Power BI Power Query*

Original file/table name	Renamed table/query name	Purpose in analysis
olist_orders_dataset	Orders	Contains order status, purchase date, approval date, and delivery dates.
olist_customers_dataset	Customers	Contains customer location fields such as city and state.
olist_order_items_dataset	Order_Items	Contains product price, freight value, product ID, seller ID, and order-item details.
olist_products_dataset	Products	Contains product category and product characteristics.
product_category_name_translation	Category_Translation	Provides English product category names.
olist_sellers_dataset	Sellers	Contains seller location information.
olist_order_reviews_dataset	Reviews	Contains review score and review information.
Final joined query	Olist_Final_Analysis	Final cleaned and joined table used for dashboard analysis.
Referenced duplicate-check query	Duplicate_Check	Used to check whether exact duplicate rows existed.

After joining the Olist tables, the final Olist_Final_Analysis dataset was cleaned and transformed in Power BI Power Query. Data types were reviewed and corrected so each field could be used properly in calculations, joins, filters, and visuals. Identifier fields such as order_id, customer_id, product_id, and seller_id were kept as text, date fields were set as Date/Time, monetary fields such as price and freight_value were set as decimal numbers, and categorical fields were kept as text.

4.2 New Analysis Fields

Several new fields were created to support the dashboard analysis. Delivery_Days measured the time between purchase and delivery, while Delay_Days compared the actual delivery date with the estimated delivery date. These were used to create Delivery_Status, which classified delivered orders as Early, On Time, or Late. Item_Total_Value, Freight_Ratio, and Purchase_YearMonth were also created to support freight analysis and monthly trend reporting.

Product categories were cleaned to improve readability. Missing categories were replaced with Unknown Category, underscores were replaced with spaces, and category names were capitalised, for example changing cool_stuff to Cool Stuff. The original Portuguese product category column was removed because the cleaned English category field was more suitable for dashboard reporting.

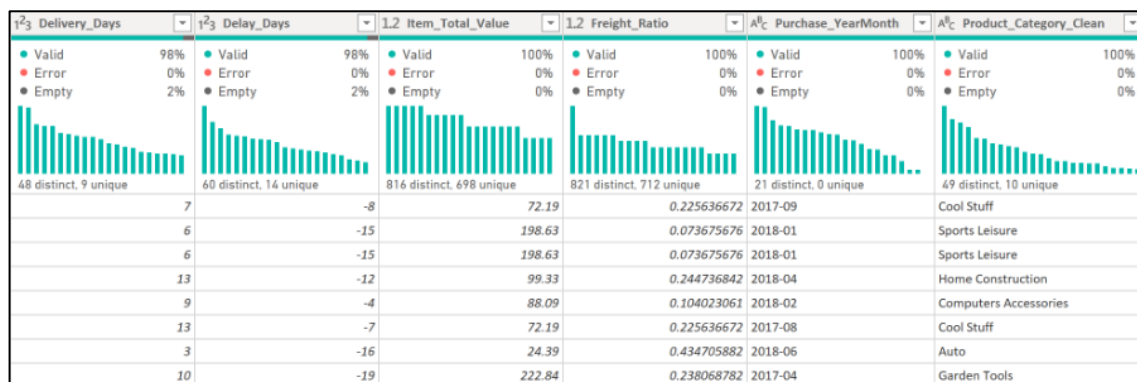


Figure 5. New analysis fields created in Power BI Power Query

4.3 Missing Values, Duplicates and Outliers

Missing values were handled through flags rather than deletion. Missing delivery dates were flagged using Delivery_Date_Status and Delivery_Data_Quality, while missing reviews were identified using Review_Status. Review scores were grouped into Review_Category. The review comment fields were removed because they were mostly blank and written in Portuguese.

Duplicate checking was completed using a referenced query called Duplicate_Check. The row count remained at 114,092, indicating that no exact duplicate rows were found. Repeated order_id values were kept because the dataset is at order-item level. Potential outliers, such as high freight ratios and long delivery durations, were flagged using High_Freight_Flag and Delivery_Duration_Flag rather than removed, as they may represent valid business cases.

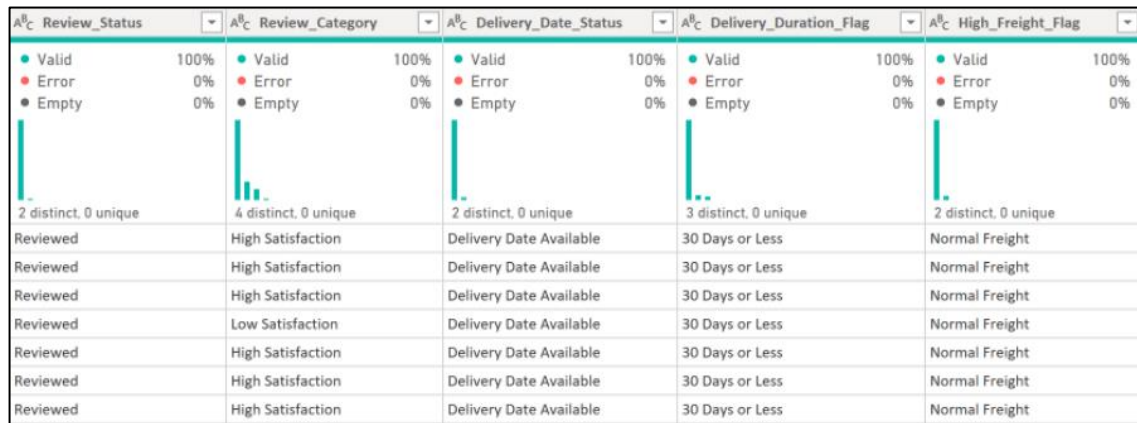


Figure 6: Data quality and flag fields used to identify review, delivery, duration, and freight issues.

7. Dashboard Development and Visualisation Design

A dashboard was developed in Power BI to present Olist's e-commerce performance in one clear executive view. It brings together sales performance, product contribution, delivery efficiency, freight cost, customer satisfaction, and regional demand on a single page.

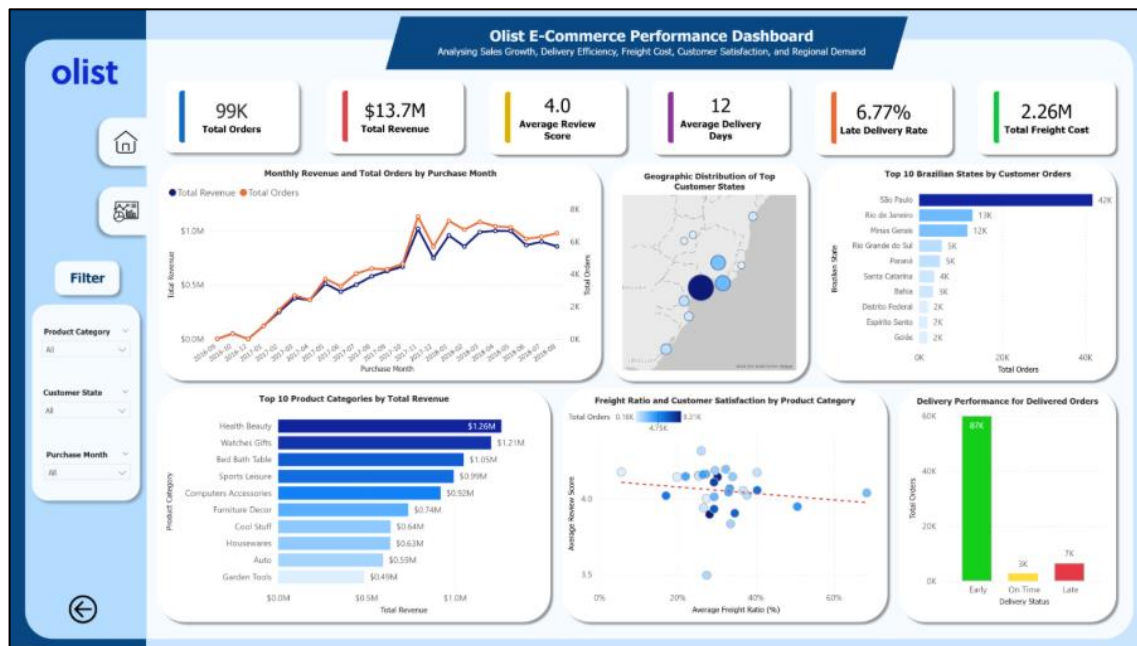


Figure 7. Final Olist E-Commerce Performance Dashboard Developed in Power BI

5.1 Dashboard Overview and KPI Cards

The top section uses KPI cards to summarise the main business indicators: 99K total orders, \$13.7M total revenue, 4.0 average review score, 12 average delivery days, 6.77% late delivery rate, and 2.26M total freight cost. These cards were placed at the top because they provide a quick snapshot of overall marketplace performance before users review the detailed visuals.



Figure 8. Power BI KPI Cards for Key Marketplace Performance Indicators

5.2 Visualisation Justification

Different visualisation types were selected based on the data being analysed and the business question each visual needed to answer. The dashboard combines KPI cards, time-series analysis, categorical comparison, relationship analysis, and geographic mapping to provide a balanced view of Olist's sales, delivery, freight, satisfaction, and regional performance.

Table 4. Visualisation Design Justification

Visual	Visual type	Justification
KPI Summary	KPI cards	Provides a quick executive overview of total orders, revenue, review score, delivery days, late delivery rate, and freight cost before users review the detailed charts
Monthly Revenue and Total Orders	Dual-axis line chart	Shows changes in revenue and order volume over time, helping users see whether revenue growth is supported by order growth
Top 10 Product Categories by Revenue	Horizontal bar chart	Compares revenue across product categories clearly. The horizontal layout keeps longer category names readable
Delivery Performance for Delivered Orders	Column chart	Compares Early, On Time, and Late delivered orders. The colour coding makes delivery performance easy to interpret
Freight Ratio and Customer Satisfaction	Scatter chart with trend line	Shows the relationship between average freight ratio and average review score, helping identify whether freight cost may be associated with satisfaction risk
Geographic Distribution of Top Customer States	Map visual	Shows where customer demand is concentrated across Brazilian states, making regional patterns easier to understand
Top 10 Brazilian States by Customer Orders	Horizontal bar chart	Supports the map by showing the exact ranking of customer order volume by state
Product Category, Customer State, Purchase Month Filters	Dropdown slicers	Allows users to filter the dashboard by product category, region, and time period while keeping the layout clean

This combination of visuals allows the dashboard to show both high-level performance and more detailed business patterns. The line chart highlights trends, bar charts support ranking and comparison while the scatter chart explores relationships, and the map communicates geographic concentration. Together, these visuals support Olist's decision-making across sales, logistics, freight management, and customer experience.

5.3 Slicers and Interactivity

The dashboard includes three dropdown slicers on the left side: Product Category, Customer State, and Purchase Month. These filters allow users to explore the dashboard by product group, region, and time period. Placing the slicers in a separate filter panel keeps the main dashboard area clean while still supporting interactive analysis.

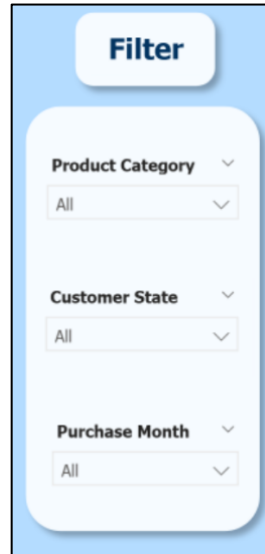


Figure 8. *Dashboard Filter Panel with Product Category, Customer State, and Purchase Month Slicers*

5.4 Dashboard Layout and Design

The dashboard uses a consistent blue colour theme with rounded visual containers, clear titles, and aligned spacing to create a professional appearance. KPI cards were placed at the top, filters were grouped on the left, and detailed visuals were arranged across the main canvas according to business themes. Meaningful colours were also used in the delivery chart, with green for early delivery, yellow for on-time delivery, and red for late delivery. Overall, the design supports both high-level monitoring and detailed exploration of Olist's sales, delivery, freight, satisfaction, and regional performance.

6. Interpretation and Insights

6.1 Audience-Tailored Communication

This analysis is primarily aimed at Olist marketplace managers and senior executives responsible for sales performance, product strategy, logistics, transport management, and customer experience. The group requires concise and actionable insights that link commercial outcomes (such as revenue and order volume) to operational factors (such as delivery performance, freight cost share, and regional demand), which enables the prioritisation of investments, the adjustment of seller and inventory policies, and the optimisation of marketing and logistics resources based on measurable KPIs. The recommendations set out below are designed for those in roles capable of driving transformation across merchandising, fulfilment, and customer experience teams.

6.2 Key Findings

The dashboard reveals six core findings that should guide strategic choices:

- (1) Olist processed 99K orders and \$13.7M in revenue as shown in KPI Cards on Figure 8.
- (2) Revenue is concentrated in Health & Beauty, Watches & Gifts, and Bed Bath Table.

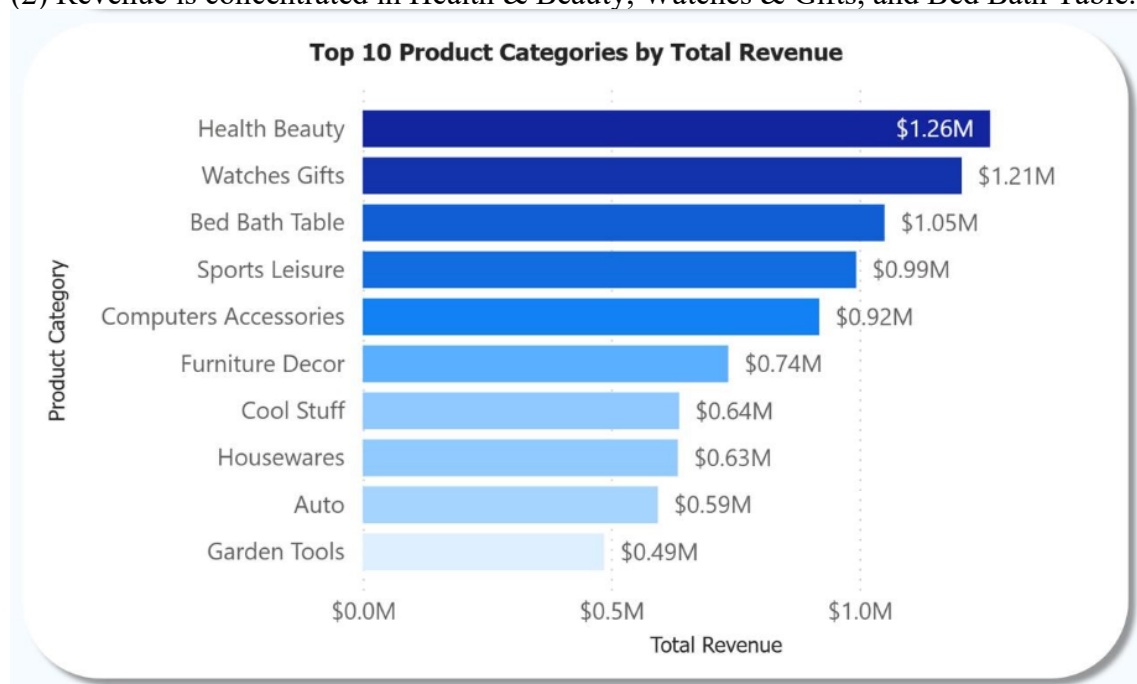


Figure 9:
Horizontal Bar Chart of Top 10 Product Categories by Revenue

(3) Platform growth accelerated through 2017 then plateaued in 2018.



Figure 10:
Line graph of Monthly Revenue and Total Orders by Purchase Month

(4) Most delivered orders arrived early or on time, while 6.77% were late.

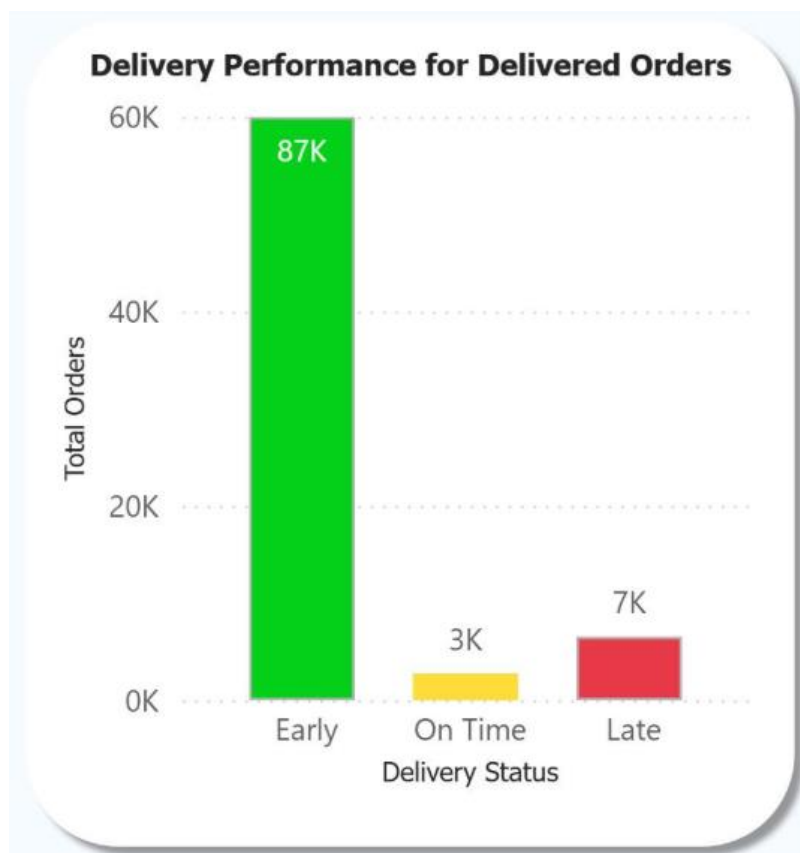


Figure 11:
Vertical Bar Chart of Delivery Performance

(5) Higher freight ratios correlate slightly with lower review scores.

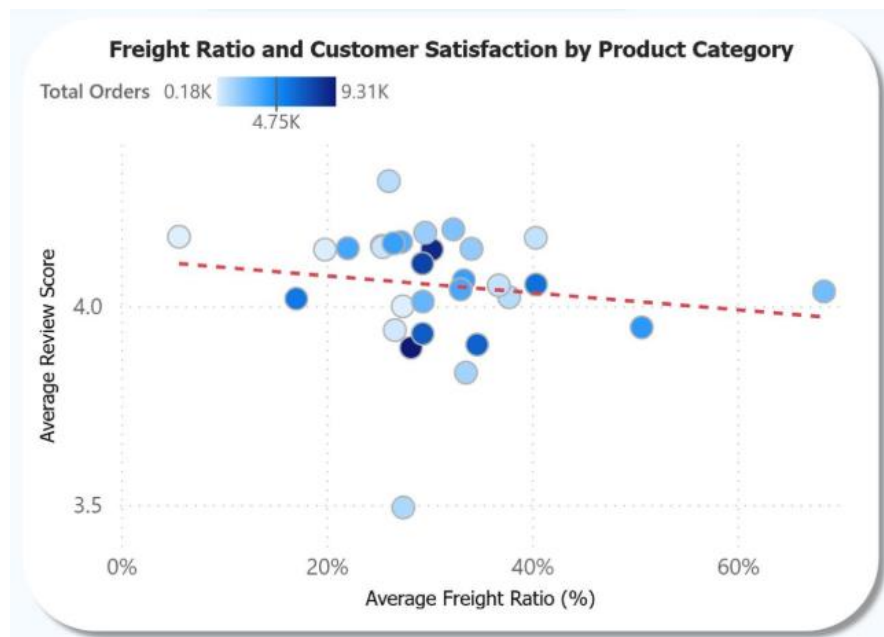


Figure 12:
Scatter Plot of Freight Ratio and Customer Satisfaction

(6) São Paulo dominates customer orders, followed by Rio de Janeiro and Minas Gerais.

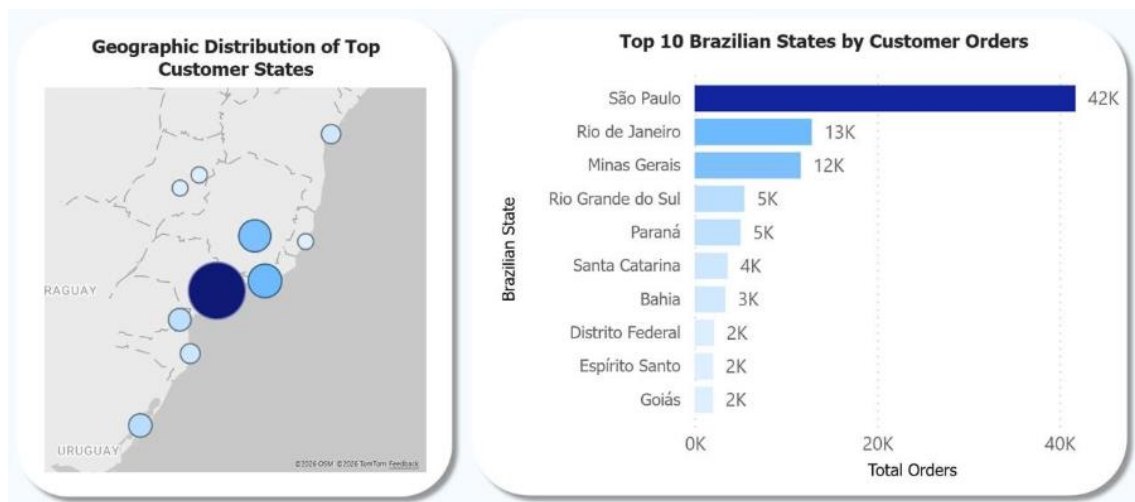


Figure 13:
Map and Bar Chart of Geographic Distribution of Customer Orders

Overall, Olist recorded 99K orders and \$13.7M in sales, demonstrating robust sales activity. The state of São Paulo accounted for an overwhelming share with 42,000 orders, more than three times that of the state in second place.

6.3 Real-World Implications

Key findings listed above suggest that current growth is driven primarily by volume rather than value, and as revenue risks are concentrated in a very small number of categories, the business would be exposed to risk should consumer preferences change or supply issues arise. For instance, the geographical concentration in São Paulo as shown in figure 13 presents opportunities for efficiency whilst also harbouring systemic risks. Prioritising logistics capacity, marketing support, and service monitoring in São Paulo could improve service efficiency because this state has the highest customer order volume. On the other hand, it increases vulnerability to local disruptions. From an operational perspective, as indicated by the 6.77% late delivery rate and the slight negative relationship between freight ratio and review score suggests that higher freight burden may affect perceived customer value.

6.4 Recommendations

To translate insight into action, implement five focused measures:

- (1) Strengthen top categories via targeted promotions, improved stock allocation, and seller enablement to protect and grow high-value revenue streams.
- (2) Concentrate logistics and marketing resources on São Paulo, Rio de Janeiro, and Minas Gerais to improve fulfillment speed and conversion.
- (3) Investigate late deliveries by seller, state, and category to identify root causes and remediate bottlenecks.
- (4) Monitor and optimise high freight-ratio categories through shipping pilots, bundling, or supplier negotiations to reduce freight share.
- (5) Track the review score alongside delivery and freight metrics as a composite KPI to measure the customer-experience impact of operational changes.

9. References

Kaggle. (n.d.). *Brazilian E-Commerce Public Dataset by Olist* [Data set].
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